

5. Evaluate $\int \frac{dx}{x^2\sqrt{x^2+4}}$.

(5 points)

3. The top and bottom margins of a poster are each 5cm and the side margins are each 4cm. If the area of the printed material on the poster is fixed at 180 square centimeters. Find the dimension of the poster with the smallest area. (4 points)

2. Find the constant k such that $f(x) = \begin{cases} -\frac{\sin kx}{x}, & x \neq 0 \\ k^3 - 4k^2 - 13k, & x = 0 \end{cases}$ will be continuous at $x = 0$. (5 points)

6. Find the area enclosed by the curves $f(x) = -x^2 + 4x$ and $g(x) = x^2 - 6x + 5$ over the interval $[0, 1]$ (5 points)

Part I. (Short answer) Write the simplified answer for the following questions in the space provided. (2 pts. each)

1. The orthogonal projection of $U = (5, 1, -3)$ along $V = (4, 4, 2)$ equals to _____
2. The volume of the parallelepiped with adjacent vectors $A(1, 2, -1)$, $B(0, -1, 1)$ and $C(1, 2, 0)$ equals to _____
3. A) $\lim_{x \rightarrow \infty} \left(\frac{x-1}{x} \right)^{3x} =$ _____
 B) $\lim_{x \rightarrow 0} \frac{\tan x - 2x}{5x - \sin x} =$ _____
4. If $f(x) = \frac{\ln x}{x}$, then the equation of the tangent line for the curve of f at $(1, 0)$ is _____
5. Let $f(x) = e^x \tan^{-1} x$. Then the equation of normal line to the graph of f at $(0, 0)$ is _____
6. Given the function $yx^2 = yx - y^3$, then $\frac{dy}{dx}$ at $(1, 1) =$ _____
7. A) $\int_x^{x^2} e^{-t} dt =$ _____
 B) $\int_{-1}^0 x e^{-x^2} dx =$ _____
8. For what value(s) of p does the improper integral $\int_1^{\infty} \frac{dx}{x^{(p-1)}}$ converge _____

Part II (Workout): Attempt all questions by showing all the necessary steps clearly on the space Provide.

1. Find the equation of the plane containing the points $A(1, 2, 3)$, $B(4, 1, -1)$ and $C(2, 2, 0)$. (5 points)

4. Given a function $f(x) = (x - 1)^2(x + 2)$.
- A) Determine the intervals where f is increasing or decreasing.
 - B) Find the relative extrema of f .
 - C) Determine the concavity of f .

(6 points)